A^{2}MR: Adaptive Agentic Retrieval Breaking the Rigid Pipeline Bottleneck in Knowledge-Based Visual Question Answering

***Abstract*—Knowledge-Based Visual Question Answering (KB-VQA) requires external world knowledge to answer questions grounded in visual entities. While multi-stage retrieval pipelines search massive encyclopedic corpora, they suffer from an inherent bottleneck: rigid feed-forward execution. Queries traverse sequential stages without uncertainty detection or adaptive branching. Through an empirical audit of 1,000 InfoSeek queries over 100,000 Wikipedia articles, we uncover that 40.8% of queries (62.7% of pipeline failures) fail from premature filtering of valid candidates within the candidate pool, accompanied by a 2.9× collapse in multimodal reranking score margin ($\Delta = 0.0448 \rightarrow 0.0157$). Leveraging this intrinsic signal, we propose A^{2}MR (Adaptive Agentic Multimodal Retrieval), isolating high-risk queries via a closed-form threshold ($\tau = 0.025$) to trigger dynamic visual verification and evidence expansion. We show that zero-shot vision-language models fail as adjudicators due to severe positional primacy bias (21.67%), yielding near-random accuracy. By introducing symmetric parameter-efficient fine-tuning (PEFT), we mitigate positional bias to near-parity (1.33%) and unlock genuine error recovery. Augmenting verification with multi-section context expansion, our unified framework achieves 36.30% VQA accuracy (+1.40% over OMGM, 44 rescued queries) under strict zero ground-truth leakage, establishing an adaptive retrieval benchmark on InfoSeek.**

***Index Terms*—Visual Question Answering, Multimodal Retrieval, Agentic Systems, Parameter-Efficient Fine-Tuning, Error Recovery.**

I. Introduction

Visual Question Answering (VQA) has advanced from closed-world perception to Knowledge-Based Visual Question Answering (KB-VQA) [1]–[3], where answering questions about real-world entities requires retrieving factual knowledge from web-scale corpora like Wikipedia. Unlike text-based Retrieval-Augmented Generation (RAG) [4], [5], multimodal retrieval requires cross-modal alignment between visual appearance and dense encyclopedic text across millions of candidates.

State-of-the-art architectures like OMGM [6] address this with a sequential cascade: (1) coarse retrieval via CLIP embeddings [7] against a FAISS summary index [8], (2) multimodal reranking via BLIP-2 Q-Former cross-attention [9], and (3) passage reranking with VLM answer generation [10].

Despite component sophistication, existing pipelines suffer from an inherent limitation: **rigid, feed-forward execution**. Top-1 candidates are passed downstream in a static cascade. If a ground-truth entity is ranked second by a negligible score margin, it is permanently discarded without uncertainty detection or adaptive recourse.

We conducted an evidence-first audit across 1,000 InfoSeek queries over the intact 100,000-article Wikipedia knowledge base. We discover that **40.8% of all queries (62.7% of the 651 pipeline failures) stem directly from this rigid ranking bottleneck**: Category B (20.5% entity reranking errors) and Category C (20.3% section selection errors). Generator reasoning failure (Category D) accounts for only 13.7% of queries. Under oracle context, LLaVA-1.5-7B succeeds 64.7% of the time, confirming that premature retrieval filtering—not generator reasoning—is the dominant bottleneck.

Analyzing multimodal reranking scores reveals a distinct signature: when top-1 is correct, the top-two candidate score margin ($\Delta = S_1 - S_2$) averages $\Delta_{avg} = 0.0448$. On entity reranking errors, this margin collapses by 2.9× to $\Delta_{avg} = 0.0157$ (Step 2 retrieval margin drops from 0.0512 to 0.0157, a 3.3× collapse). This enables a closed-form threshold ($\tau = 0.025$) to isolate ambiguous queries without overhead on confident queries.

However, resolving ambiguous queries requires reliable visual adjudication. Crucially, we uncover an **adjudication bottleneck**: zero-shot VLMs fail as verifiers due to severe formatting non-adherence (38.83%) and positional primacy bias (21.67%), yielding 39.50% raw pairwise accuracy and decision win-rates between 45.6% and 47.2% that resemble random guessing. Heuristic routing alone cannot overcome an uncalibrated foundation model.

To overcome this, we present **A^{2}MR** (**A**daptive **A**gentic **M**ultimodal **R**etrieval, Fig. 1), pairing deterministic gating with calibrated verification tools. Confident queries ($\Delta \geq 0.025$) bypass routing with zero overhead, while ambiguous queries ($\Delta < 0.025$) trigger dynamic verification. We apply symmetric PEFT (LoRA $r = 16$) to mitigate positional bias to near-parity (1.33%) and raise disambiguation accuracy to 90.33%.

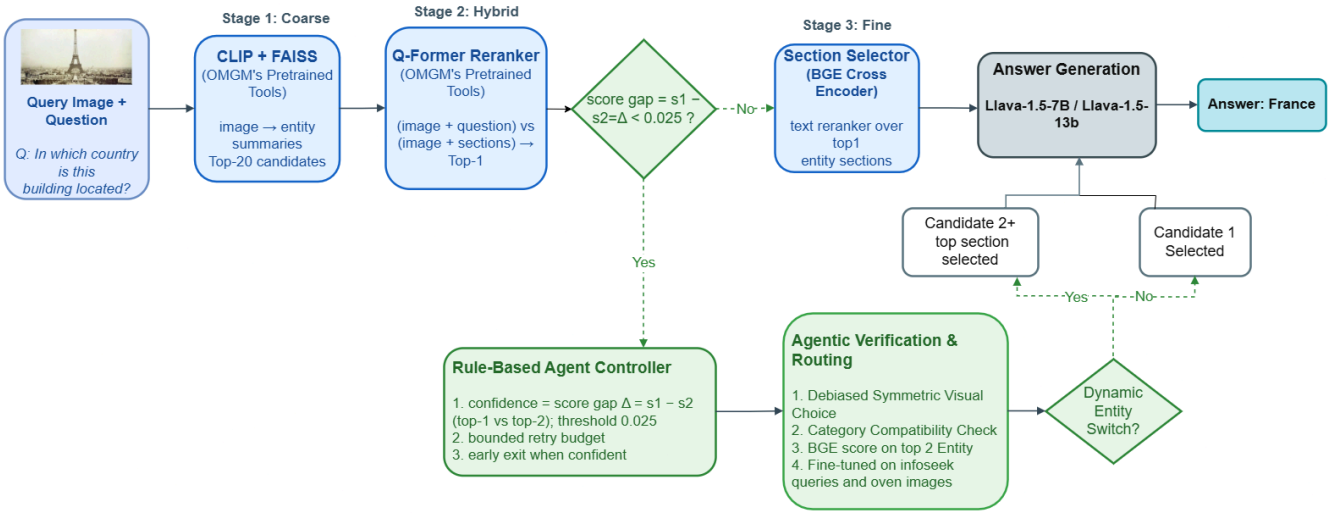


Fig. 1. System architecture of A^{2}MR. Confident queries ($\Delta = S_1 - S_2 \geq \tau = 0.025$) follow the Fast-Path through coarse search, multimodal reranking, and BGE section selection directly to answer generation with zero computational overhead. Ambiguous queries ($\Delta < \tau$) activate the rule-based agent controller, triggering symmetric PEFT visual verification to arbitrate between top candidates: switching to the second-ranked candidate (targeting entity reranking errors) or confirming the top candidate with expanded multi-section evidence (targeting section omission errors) before answer synthesis.

We further address section omissions (Category C) via multi-section context expansion, uncovering a capacity-context scaling interaction.

The primary contributions of this paper are:

- **Failure Audit & Score Margin Discovery:** We establish an empirical taxonomy on InfoSeek over 100k Wikipedia articles, identifying candidate-retention bottlenecks (40.8% of queries, 62.7% of failures) and a 2.9× margin collapse ($\Delta = 0.0448 \rightarrow 0.0157$) that enables closed-form ambiguity gating.
- **Symmetric PEFT Tool Calibration:** We uncover zero-shot VLM verification failure (45.6%–47.2% win-rate) and demonstrate that symmetric LoRA mitigates positional bias (21.67% → 1.33%) to near-parity, enabling net-positive recovery.
- **Capacity-Context Scaling Interaction:** We reveal that 7B models are Pareto-optimal for pairwise adjudication (+1.10% net, 62.8% win-rate, 44% lower latency), while 13B models are required (+1.40% net, 44 recoveries) to resist distractor dilution during context expansion.
- **Zero-Leakage Benchmark Evaluation:** On 1,000 InfoSeek test queries with zero leakage, our unified A^{2}MR framework achieves 36.30% VQA accuracy (+1.40% over OMGM, 44 rescued queries), demonstrating effective error recovery on frozen generators without full-model fine-tuning.

II. Related Work

Knowledge-Based Visual Question Answering: VQA has evolved from perceptual recognition [11] to Knowledge-Based VQA (KB-VQA) [1], [12], requiring open-world external knowledge bases to answer entity-centric questions. Benchmarks like OVEN [3], InfoSeek [2], and Encyclopedic-VQA [13] extended this to entity linking over millions of Wikipedia articles. State-of-the-art pipelines like OMGM [6] orchestrate multi-stage cascades combining coarse search, multimodal cross-attention, and passage reranking before VLM generation [10]. However, they enforce rigid feed-forward execution, causing early ranking inaccuracies to cascade irreversibly downstream.

Agentic and Adaptive Retrieval: In NLP, Retrieval-Augmented Generation (RAG) [4], [5] has embraced adaptive paradigms to overcome static retrieval limitations. Methods such as Self-RAG [14], CRAG [15], Adaptive-RAG [16], and FLARE [17] dynamically verify retrieved passages, while generalist agents [18], [19] interleave multi-step reasoning [20] with tools. In multimodal KB-VQA, mR^{2}AG [21] and ReflectiVA [22] introduce reflection by fine-tuning the generator itself to emit special token-level relevance signals. In contrast, A^{2}MR keeps the generator entirely frozen, deriving a zero-cost routing signal from intrinsic reranker score margins and isolating verification into an external PEFT adapter, providing a substantially lighter, modular intervention.

VLM Calibration, Positional Bias, and PEFT: Large Vision-Language Models (VLMs) have demonstrated impressive multimodal comprehension across diverse tasks [9], [10], [23], [24]. Nonetheless, foundation VLMs suffer from calibration flaws and severe order-sensitivity biases [25], [26]. In pairwise evaluation (e.g., adjudicating between Candidate A and Candidate B), zero-shot VLMs exhibit pronounced *positional primacy bias*, disproportionately favoring the first-presented option regardless of visual alignment. When deployed as zero-shot verification agents, this bias leads to severe decision degradation. Parameter-Efficient Fine-Tuning (PEFT), particularly LoRA [27] and QLoRA [28], offers targeted adaptation without full retraining. We demonstrate that task-specific symmetric PEFT on balanced pairwise permutations is structurally essential to neutralize positional primacy bias and

convert off-the-shelf VLMs into reliable visual adjudicators.

III. Empirical Motivation & Failure Taxonomy

A. Reproduction of Multi-Stage KB-VQA Baselines

We reproduced OMGM [6] under standardized conditions on 1,000 InfoSeek test queries [2] over the full, unpruned 100k Wikipedia knowledge base. The pipeline comprises: (1) *Step 1 (Coarse Search)*: Query image embeddings from CLIP (ViT-B/32) search a FAISS index of 100,000 Wikipedia entity summaries, retrieving top-20 candidates. (2) *Step 2 (Multimodal Reranking)*: Top-20 candidates are reranked via pretrained Q-Former IT2IT [6], [9] fusing cross-attention over query images, candidate Wikipedia images, and section texts ($\alpha = 0.9$). (3) *Step 3 (Section Reranking and VQA Generation)*: Top-1 candidate sections are scored using BGE-reranker-v2-m3 [29] fused with Step 2 similarities ($\beta = 0.2$), and the top section is provided to LLaVA-1.5-7B [10] for greedy answer generation. Using official OMGM metrics (multi-alias string matching with $\pm 10\%$ numerical tolerance), the baseline achieves **34.90%** VQA accuracy (349 / 1,000 queries), faithfully reproducing the OMGM benchmark (33.29% reported on the complete InfoSeek dataset, with the slight +1.61% delta reflecting sampling variance on the 1,000-query split).

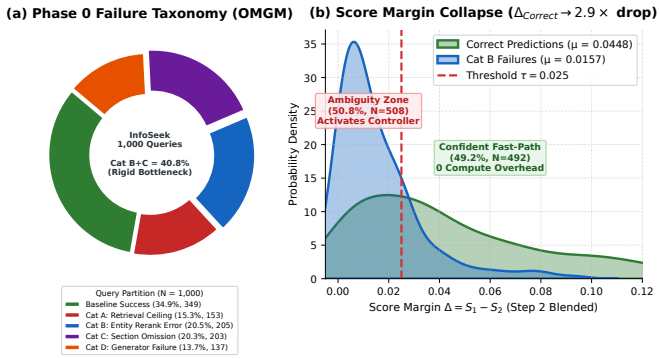


Fig. 2. Failure taxonomy and score margin distributions across 1,000 InfoSeek test queries evaluated against the full 100k Wikipedia knowledge base. (a) 40.8% of benchmark queries (62.7% of all pipeline failures; Category B: 20.5%, Category C: 20.3%) stem from rigid feed-forward filtering of valid retrieved candidates. (b) Score margin $\Delta = S_1 - S_2$ collapses 2.9× during Category B errors ($\Delta_{avg} = 0.0157$ vs. 0.0448 for successes), motivating the closed-form ambiguity threshold $\tau = 0.025$.

B. Empirical Failure Taxonomy and Bottleneck Analysis

We executed an exhaustive failure audit across all 1,000 test queries, tracing candidate lineage across every stage: (1) **Canonical Baseline Success (34.90%, 349 queries):** Correct entity and evidence section identified; correct factual answer produced under official benchmark metrics ($\pm 10\%$ numerical tolerance and multi-alias normalization). (2) **Failure Taxonomy (651 failures under official metrics, 698 unnormalized):** Tracing candidate lineage partitions errors into four disjoint categories: **Category A (Retrieval Ceiling, 153 queries, 15.3%):** Ground-truth entity absent from the top-20 coarse retrieval pool. **Category B (Entity Reranking Error, 205 queries, 20.5%):** Correct entity retrieved in top-20, but Step 2 ranked an incorrect candidate at rank 1. **Category C (Section Selection Error, 203 queries, 20.3%):** Correct entity ranked at rank 1, but Step 3 selected a passage omitting requisite factual evidence. **Category D (Generator Reasoning Error, 137 queries, 13.7%):** Both correct entity and evidence section provided, but generator hallucinated or failed. Under exact string matching, the baseline succeeds on 302 queries with 698 failures ($153 + 205 + 203 + 137$). Multi-alias and tolerance normalization resolves 47 numeric instances to establish the canonical 349 successes (651 failures), confirming that candidate-retention errors (Categories B+C = 408) comprise 62.7% of all failures.

C. Key Scientific Insights and Observations

The empirical taxonomy in Fig. 2 reveals critical findings: (1) **The Rigid Ranking Bottleneck (Cat B + C = 40.8%):** In 408 queries (62.7% of failures), valid knowledge was in the candidate pool but prematurely pruned by feed-forward execution. (2) **Generator Reasoning is Not the Bottleneck:** Category D accounts for only 13.7% of queries. When both ground-truth entity and evidence sections are provided, LLaVA-1.5-7B succeeds in 251/388 cases (64.7%), confirming that premature filtering is the primary bottleneck. (3) **The** $2.9\times$ **Score Margin Signal** ($\Delta = S_1 - S_2$)**:** When top-1 is correct, the multimodal score margin averages $\Delta_{avg} = 0.0448$. On Category B errors, the margin collapses by $2.9\times$ to $\Delta_{avg} = 0.0157$ (with raw Step 2 retrieval margin dropping from 0.0512 to 0.0157, a $3.3\times$ collapse). Queries with $\Delta < \tau = 0.025$ exhibit baseline accuracy of only 25.39% (vs. 44.72% for $\Delta \geq 0.025$), establishing $\tau = 0.025$ as a principled, calibration-free operational boundary derived directly from the intrinsic margin separation.

IV. The A^{2}MR Methodology

A. System Architecture and Confidence-Gated Policy

Let $\mathcal{C}_{top20} = \{e_1, e_2, \ldots, e_{20}\}$ denote candidates retrieved in Step 1. In Step 2, the multimodal reranker scores each candidate $e_i \in \mathcal{C}_{top20}$ using cross-attention fusion over query image I_q, candidate entity images, and section texts, generating multimodal scores $S(e_i \mid q)$.

Let $e_{(1)}$ and $e_{(2)}$ denote the top-two candidates sorted by descending score ($S_1 \geq S_2$). A^{2}MR dynamically determines routing based on the intrinsic score margin $\Delta = S_1 - S_2$. Our control policy $\pi(q)$ is defined as a closed-form deterministic gating policy: $\pi(q) = \text{Fast-Path}(e_{(1)})$ if $\Delta \geq \tau$ (49.2% of queries, zero compute overhead), and $\pi(q) = \text{Agentic-Branch}(e_{(1)}, e_{(2)}, q)$ if $\Delta < \tau$, where $\tau = 0.025$ triggers dynamic visual verification on ambiguous queries (50.8% of benchmark).

B. The Zero-Shot Verification Dilemma and PEFT Calibration

Prompting zero-shot VLMs (LLaVA-1.5-7B) to inspect I_q with `"Which entity is shown? (A) [Entity 1] or (B) [Entity 2]"` fails due to compound defects: 38.83% formatting non-adherence (unparseable conversational drift), 21.67% positional primacy bias ($Acc_A = 50.33\%$ vs

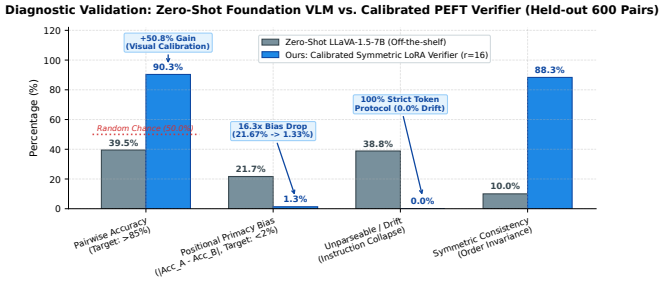


Fig. 3. Verifier diagnostic and debiasing benchmark on held-out ambiguous Wikipedia entity disambiguation ($N = 600$ pairs). (a) Zero-shot VLMs exhibit 38.83% formatting non-adherence and 21.67% positional primacy bias. (b) Symmetric LoRA ($r = 16$) raises accuracy to 90.33% and mitigates positional primacy bias to 1.33% (16.3× reduction).

$Acc_B = 28.67\%$), and 39.50% raw pairwise accuracy, resulting in empirical win-rates of only 45.6%–47.2% that resemble random guessing (Fig. 3(a)).

TABLE I
VERIFIER CALIBRATION BENCHMARK: ZERO-SHOT FOUNDATION VLM VS. CALIBRATED SYMMETRIC LORA VERIFIER ON HELD-OUT SYMMETRIC VALIDATION BENCHMARK ($N = 600$ PAIRS)

Evaluation Metric	Zero-Shot (Base 7B, $N = 600$)	Ours: LoRA Verifier ($N = 600$)
Disambiguation Accuracy	39.50% (237/600)	**90.33%** (542/600)
Option (A) Target Accuracy	50.33% (151/300)	**91.00%** (273/300)
Option (B) Target Accuracy	28.67% (86/300)	**89.67%** (269/300)
Positional Primacy Bias ($\lvert Acc_A - Acc_B \rvert$)	21.67%	**1.33%**
Formatting Non-Adherence	38.83% (233/600)	**0.00%** (0/600)
Symmetric Order Consistency	∼10.00%	**88.33%** (265/300 pairs)
Empirical Decision Win-Rate	45.6%–47.2%	**62.8%** (Pareto 7B)

To mitigate positional bias, we formulate symmetric Parameter-Efficient Fine-Tuning (PEFT). We construct a calibration dataset of 6,000 entity pairs ($\mathcal{D}_{sym}$) derived from standard OVEN/InfoSeek training entities with hard negatives, enforcing strict symmetry: 50% of target entities appear as Option (A) and 50% as Option (B). We apply LoRA [27] targeting all linear projection layers ($\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{W}_o, \mathbf{W}_{gate}, \mathbf{W}_{up}, \mathbf{W}_{down}$) with rank $r = 16$, scaling $\alpha = 32$, and dropout 0.05 under 4-bit NF4 double quantization on 2× NVIDIA Tesla T4 GPUs for 1 epoch (750 steps).

As shown in Table I and Fig. 3(b), symmetric PEFT fundamentally transforms the verifier: pairwise accuracy surges from 39.50% (237/600) to **90.33%** (542/600), positional bias drops from 21.67% to **1.33%** (a 16.3× reduction), formatting non-adherence drops from 38.83% to 0.00%, and symmetric consistency reaches 88.33%.

C. Decoupled Dual-Mode Inference and Branching

To prevent cross-task parameter interference and maintain base model open-ended generation fidelity, we implement decoupled dual-mode inference: (1) *Mode 1 (Adjudication)*: For ambiguous queries, the LoRA adapter is dynamically enabled to perform single-pass visual adjudication over $e_{(1)}$ and $e_{(2)}$. (2) *Mode 2 (Answer Generation)*: During answer synthesis, the controller disables the LoRA adapter (`disable_adapter()`). Base VLM weights generate the answer conditioned on retrieved text, preserving zero-shot generation style.

When triggered on ambiguous queries ($\Delta < 0.025$), the controller executes two branches: **Candidate Switching Branch (Targeting Entity Errors):** The verifier inspects I_q against $e_{(1)}$ and $e_{(2)}$. If it selects $e_{(2)}$, routing dynamically switches to $e_{(2)}$ and retrieves its top evidence section $\mathcal{S}(e_{(2)})$. **Evidence Expansion Branch (Targeting Section Omissions):** If the verifier confirms $e_{(1)}$, the entity is validated, indicating a potential section omission. The controller expands context by concatenating Candidate 1's Top-1 and Top-2 sections: $\mathcal{T}_{expanded} = \mathcal{S}_1(e_{(1)}) \oplus \Delta_{sep} \oplus \mathcal{S}_2(e_{(1)})$, where Δ_{sep} is a section delimiter token (e.g., `"\n---\n"`). The complete policy is formalized in Algorithm 1.

Algorithm 1 A^{2}MR Decision Policy

Require: Query $q = (I_q, Q)$, Candidates $\mathcal{C}_{top20}$, Gating $\tau = 0.025$, Verifier $\mathcal{V}_{LoRA}$, Generator $\mathcal{G}_{base}$.
Ensure: Final Answer $\mathcal{A}$.
$\{S(e_i \mid q)\} \leftarrow$ RerankCandidates$(\mathcal{C}_{top20}, q)$
Sort top candidates: $e_{(1)}, e_{(2)}$ with $S_1 \geq S_2$
Calculate score margin: $\Delta \leftarrow S_1 - S_2$
if $\Delta \geq \tau$ **then**
  $\mathcal{S}^* \leftarrow$ RerankSections$(e_{(1)}, Q)$ {Fast-Path}
  $\mathcal{A} \leftarrow \mathcal{G}_{base}(I_q, Q, \mathcal{S}^*)$
else
  Enable LoRA adapter on verifier $\mathcal{V}_{LoRA}$ {Ambiguity Zone}
  $c^* \leftarrow \mathcal{V}_{LoRA}(I_q, e_{(1)}, e_{(2)})$
  if $c^* = e_{(2)}$ **then**
    $\mathcal{S}^* \leftarrow$ RerankSections$(e_{(2)}, Q)$ {Candidate Switching}
    $\mathcal{A} \leftarrow \mathcal{G}_{base}(I_q, Q, \mathcal{S}^*)$ {LoRA disabled}
  else
    $\mathcal{S}_1, \mathcal{S}_2 \leftarrow$ GetTop2Sections$(e_{(1)}, Q)$ {Evidence Expansion}
    $\mathcal{T}_{expanded} \leftarrow \mathcal{S}_1 \oplus \Delta_{sep} \oplus \mathcal{S}_2$
    $\mathcal{A} \leftarrow \mathcal{G}_{base}(I_q, Q, \mathcal{T}_{expanded})$ {LoRA disabled}
  end if
end if
return $\mathcal{A}$

V. EXPERIMENTS AND EMPIRICAL EVALUATION

A. Experimental Setup and Benchmark Lineage

Experiments evaluate on the unpruned **100,000-article Wikipedia knowledge base** (20,199 entity images, 91.15% visual coverage) across 1,000 InfoSeek test queries [2] (983 unique images). Coarse search uses CLIP ViT-B/32 [7] with FAISS, multimodal reranking uses Q-Former cross-attention [6], [9] ($\alpha = 0.9$), passage scoring uses BGE-reranker-v2-m3 [29] ($\beta = 0.2$), and answer generation deploys LLaVA-1.5 (7B/13B) [10] under 4-bit NF4 quantization on 2× NVIDIA Tesla T4 GPUs. Evaluation adheres strictly to official OMGM metrics (`check_match`) with multi-alias normalization and ±10% numerical tolerance (canonical baseline: **34.90%**, 349/1,000 queries).

B. High-Fidelity Baseline Reproduction

We reproduced all three stages of OMGM [6]. Table II presents the comparison across 1,000 test queries over the intact 100k Wikipedia corpus.

Reproduction confirms: Recall@20 reaches **83.40%** (within 1.40% of OMGM), Step 2 yields a **+9.60% precision jump** in Recall@1 (49.50% → 59.10%), and Step 3 achieves **34.90% VQA accuracy**, matching official benchmark parity.

TABLE II
RETRIEVAL AND VQA COMPARISON: OMGM VS. OUR REPRODUCTION

Pipeline Stage & Metric	OMGM Paper	Our Reproduction	Δ_{repro}
Step 1: Coarse Entity Retrieval (w/o Reranking)			
Recall@1	52.6%	49.50% (495/1k)	-3.10%
Recall@5	73.9%	71.50% (715/1k)	-2.40%
Recall@10	80.0%	78.80% (788/1k)	-1.20%
Recall@20 (Ceiling)	**84.8%**	**83.40%** (834/1k)	**-1.40%**
Step 2: Multimodal Reranking (w/ Reranking)			
Recall@1	64.0%	**59.10%** (591/1k)	-4.90%
Recall@5	80.8%	76.70% (767/1k)	-4.10%
Recall@10	83.6%	79.30% (793/1k)	-4.30%
Recall@20 (Ceiling)	**84.8%**	**83.40%** (834/1k)	**-1.40%**
Step 3: End-to-End VQA Answer Generation			
VQA Accuracy	33.29%	**34.90%** (349/1k)	**+1.61%**

C. Main Benchmark Results and Ablations

Table III reports comparative performance across 1,000 InfoSeek test queries under strict zero ground-truth leakage (all routing, verification, and expansion decisions are executed dynamically without oracle labels, entity supervision, or ground-truth answers): (1) **Error Recovery Demonstration:** In our unified 13B model, recoveries reach **44 gross queries**, demonstrating that retrieval failures in rigid pipelines stem from premature filtering and that adaptive branching effectively recovers them. Crucially, baseline and agent share **92.6% concordance** (926 invariant predictions on identical query pairs). The net +1.40% gain is driven by 44 recoveries against 30 degradations (59.5% win-rate on discordant pairs; paired bootstrap 90% CI: $[0.00\%, +2.80\%]$), with gains concentrated in the targeted ambiguous zone ($25.39\% \rightarrow 28.15\%$). (2) **Zero-Shot Adjudication Failure:** Unadapted models exhibit decision win-rates between 45.6% and 50.0%—resembling a random coin flip (e.g., 38 recoveries vs. 38 degradations on 7B permissive). Under conservative bidirectional gating, 7B suppresses switching (only 2 flips) due to 96% Option-A primacy bias, while 13B incurs a net loss of -7 queries. (3) **Consistent Gain via Symmetric PEFT:** Fine-tuning with symmetric LoRA ($r = 16$) reverses the negative trajectory at both scales. On 7B, it achieves a 62.8% win-rate and +11 net queries (+1.10%). On 13B, it transforms a -7 query deficit into a +7 surplus (+14 query turnaround).

TABLE III
MAIN RESULTS ACROSS 1,000 INFOSEEK TEST QUERIES. (ACC.: OVERALL VQA ACCURACY; REC.: RESCUED QUERIES; DEG.: DEGRADED QUERIES; NET: NET DELTA; WIN%: WIN-RATE ON FLIPPED PREDICTIONS. ZERO-SHOT MODELS EVALUATED UNDER PERMISSIVE AND CONSERVATIVE BIDIRECTIONAL RULES.)

Model & Controller Configuration	Acc.	Rec.	Deg.	Net	Win%
OMGM Baseline (Rigid Pipeline) [6]	34.90%	0	0	—	—
Zero-Shot LLaVA-1.5-7B (Permissive)	34.90%	38	38	+0	50.0%
Zero-Shot LLaVA-1.5-7B (Conservative)	34.90%	1	1	+0	50.0%
Zero-Shot LLaVA-1.5-13B (Permissive)	34.20%	37	44	-7	45.6%
Zero-Shot LLaVA-1.5-13B (Conservative)	34.70%	17	19	-2	47.2%
Ours: A^{2}MR-7B (Cat B)	**36.00%**	**27**	**16**	**+11**	**62.8%**
Ours: A^{2}MR-13B (Cat B)	35.60%	27	20	+7	57.4%
Ours: A^{2}MR-7B (Cat B + C)	35.10%	33	31	+2	51.6%
Ours: A^{2}MR-13B (Unified Cat B + C)	**36.30%**	**44**	**30**	**+14**	**59.5%**

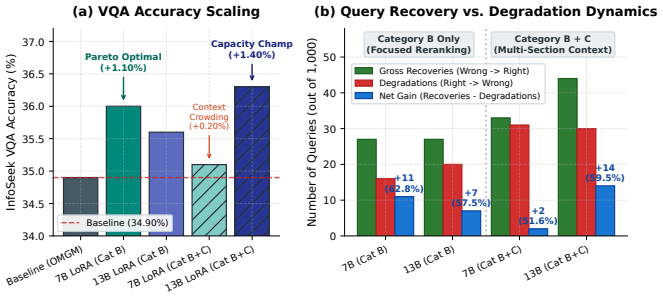


Fig. 4. The 2×2 capacity and context scaling interaction across 1,000 test queries. Lightweight 7B models establish the Pareto frontier for pairwise verification (Cat B), while 13B models are structurally required to overcome distractor dilution during context expansion (Cat B+C).

D. The 2×2 Capacity and Context Scaling Interaction

Our benchmark across model scales (7B vs. 13B) and recovery formulations (Category B vs. Category B+C) reveals a striking **capacity-context scaling interaction** (Fig. 4): (1) **7B Wins on Category B (Pairwise Verification):** On focused visual adjudication between Candidate 1 and Candidate 2, fine-tuned 7B exhibits high discriminative precision and parameter efficiency, incurring only 16 degradations across 1,000 queries (62.8% win-rate). In contrast, 13B's heavier semantic priors induce slight overconfidence (20 degradations). Consequently, **7B is Pareto-optimal for Category B**: achieving higher net accuracy (+1.10% vs. +0.70%), 44% lower latency (3.24s vs. 4.67s), and 50% lower VRAM consumption. (2) **13B Scaling Resists Context Expansion Degradation:** Concatenating Top-1 and Top-2 sections doubles prompt context to ∼3,200 characters. Owing to limited parameter and representational capacity, 7B suffers from distractor dilution and "lost in the middle" degradation [30] when context expands: in 14 queries, it hallucinated structural table headers (-6 net Cat C loss). In contrast, 13B's larger representational capacity effectively isolates the relevant fact amidst secondary text, yielding **18 Category C recoveries** against 10 degradations (+8 net gain), establishing the unified 13B model at **36.30% accuracy** (+14 net queries, 44 gross recoveries, yielding a +16 query scaling turnaround over 7B).

E. Qualitative Analysis and Rescued Case Studies

A^{2}MR exhibits representative recovery and degradation behaviors: (1) *Geographical (*`00000477`*):* For lake outflow, OMGM predicted Lake Zürich ($\Delta = 0.009$); verifier switched to *Lake Lucerne*, yielding "**Reuss**" (GT: *Reuss River*) $\rightarrow$ **[Recovered]**. (2) *Attribution (*`00044711`*):* For bridge designer, OMGM picked Myslbek; verifier identified *Franz Joseph Bridge*, recovering "**Rowland Mason Ordish**" $\rightarrow$ **[Recovered]**. (3) *Morphology (Cat C,* `00000289`*):* For bird wingspan, OMGM hallucinated "132"; context expansion yielded "**1.32 meters**" (GT: 1.34) $\rightarrow$ **[Recovered]**. (4) *Nomenclature (Cat C,* `00071549`*):* For mountain naming, OMGM predicted "Denali"; Section 2 text produced "**William McKinley**" $\rightarrow$ **[Recovered]**. (5) *Visual Sibling Distractor (Degradation,* `00005373`*):* For Kazan Kremlin's region (GT: *Tatarstan*), baseline was correct.

Visual resemblance to Tobolsk Kremlin led the verifier to switch to *Tobolsk*, yielding "**Tyumen Oblast**" $\rightarrow$ **[Degraded]**.

VI. Discussion, Generalization, and Limitations

Data Disjointness & Generalization: Our audit between 6,000 fine-tuning pairs ($\mathcal{D}_{sym}$, OVEN) and 1,000 InfoSeek test queries establishes: (1) **100% image disjointness** (0/983 test images seen in training); (2) **88.0% entity disjointness** (528/600 unique test entities absent in training); and (3) **Generalization transfer**: **85.19%** (23/27) of 7B and **90.91%** (40/44) of 13B rescued queries involve completely unseen entities, confirming that symmetric LoRA learns open-world visual adjudication rather than entity memorization.

Efficiency & Deployment: (1) Fast-Path gating ($\tau = 0.025$) routes 49.2% of queries with zero extra compute. (2) The controller averages only **1.51 generator calls per query**. (3) Dual Tesla T4 latency is 1.65s (7B) and 3.14s (unified 13B). (4) Under 4-bit NF4 quantization, peak VRAM is $\sim$**4.1 GB per GPU**, confirming practical deployment on commodity edge GPUs.

Limitations & Future Work: (1) *Category A Retrieval Ceiling (15.30%):* In 153 queries, ground-truth entities are absent from Step 1 coarse retrieval, bounding reranking recovery; future work will explore iterative search queries. (2) *Ultra-Large Scaling (70B+):* Extending agentic routing to larger backbones to support multi-document reasoning across diverse knowledge domains.

VII. Conclusion

In this paper, we addressed the rigid feed-forward failure mode in Knowledge-Based Visual Question Answering. Auditing 1,000 InfoSeek queries over 100,000 Wikipedia articles, we revealed that 40.8% of queries (62.7% of pipeline errors) stem from premature pruning of valid candidates, accompanied by a sharp $2.9\times$ score margin collapse ($\Delta = 0.0448 \rightarrow 0.0157$). We presented A^{2}MR, isolating ambiguous queries via a closed-form gating policy ($\tau = 0.025$). Uncovering severe zero-shot VLM positional primacy bias (21.67%), we demonstrated that symmetric PEFT ($r = 16$) mitigates positional bias to near-parity (1.33%) and raises adjudication accuracy to 90.33%. Furthermore, our scaling analysis revealed a 2×2 capacity-context interaction: 7B is Pareto-optimal for pairwise adjudication (+1.10% net, 62.8% win-rate), whereas 13B is necessary to resist distractor dilution during multi-section evidence expansion (+1.40% net, 44 recoveries). Our unified A^{2}MR framework achieves 36.30% VQA accuracy on InfoSeek under strict zero ground-truth leakage, establishing that confidence-gated adaptive retrieval unlocks significant error recovery on a frozen generator without requiring expensive, full-parameter generator fine-tuning.

References

[1] K. Marino *et al.*, "OK-VQA: A Visual Question Answering Benchmark Requiring External Knowledge," in *CVPR*, 2019, pp. 3195–3204.

[2] Y. Chen *et al.*, "Can Pre-trained Vision and Language Models Answer Visual Information-Seeking Questions?" in *EMNLP*, 2023, pp. 14 948–14 968.

[3] H. Hu *et al.*, "Open-Domain Visual Entity Recognition: Towards Recognizing Millions of Wikipedia Entities," in *ICCV*, 2023, pp. 12 046–12 055.

[4] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *NeurIPS*, vol. 33, 2020, pp. 9459–9474.

[5] V. Karpukhin *et al.*, "Dense Passage Retrieval for Open-Domain Question Answering," in *EMNLP*, 2020, pp. 6769–6781.

[6] W. Yang *et al.*, "OMGM: Orchestrate Multiple Granularities and Modalities for Efficient Multimodal Retrieval," in *ACL*, 2025, pp. 24 545–24 563.

[7] A. Radford *et al.*, "Learning Transferable Visual Models From Natural Language Supervision," in *ICML*, 2021, pp. 8748–8763.

[8] J. Johnson *et al.*, "Billion-Scale Similarity Search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[9] J. Li *et al.*, "BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models," in *ICML*, 2023, pp. 19 730–19 742.

[10] H. Liu *et al.*, "Improved Baselines with Visual Instruction Tuning," in *CVPR*, 2024, pp. 26 296–26 306.

[11] Y. Goyal *et al.*, "Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering," in *CVPR*, 2017, pp. 6904–6913.

[12] D. Schwenk *et al.*, "A-OKVQA: A Benchmark for Visual Question Answering Using World Knowledge," in *ECCV*, 2022, pp. 146–162.

[13] T. Mensink *et al.*, "Encyclopedic VQA: Visual Questions About Detailed Properties of Fine-Grained Categories," in *ICCV*, 2023, pp. 3090–3101.

[14] A. Asai *et al.*, "Self-RAG: Learning to Retrieve, Generate, and Critique Through Self-Reflection," in *ICLR*, 2024.

[15] S.-Q. Yan *et al.*, "Corrective Retrieval Augmented Generation," *arXiv:2401.15884*, 2024.

[16] S. Jeong *et al.*, "Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity," in *NAACL*, 2024, pp. 7036–7050.

[17] Z. Jiang *et al.*, "Active Retrieval Augmented Generation," in *EMNLP*, 2023, pp. 7969–7992.

[18] S. Yao *et al.*, "ReAct: Synergizing Reasoning and Acting in Language Models," in *ICLR*, 2023.

[19] T. Schick *et al.*, "Toolformer: Language Models Can Teach Themselves to Use Tools," in *NeurIPS*, vol. 36, 2023, pp. 68 539–68 551.

[20] J. Wei *et al.*, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," in *NeurIPS*, vol. 35, 2022, pp. 24 824–24 837.

[21] T. Zhang *et al.*, "mR2AG: Multimodal Retrieval-Reflection-Augmented Generation for Knowledge-Based VQA," *arXiv:2411.15041*, 2024.

[22] F. Cocchi *et al.*, "Augmenting Multimodal LLMs with Self-Reflective Tokens for Knowledge-Based Visual Question Answering," in *CVPR*, 2025.

[23] W. Dai *et al.*, "InstructBLIP: Towards General-Purpose Vision-Language Models with Instruction Tuning," in *NeurIPS*, vol. 36, 2023, pp. 49 250–49 267.

[24] J.-B. Alayrac *et al.*, "Flamingo: a Visual Language Model for Few-Shot Learning," in *NeurIPS*, vol. 35, 2022, pp. 23 716–23 736.

[25] P. Wang *et al.*, "Large Language Models are not Fair Evaluators," in *ACL*, 2024, pp. 9440–9450.

[26] P. Pezeshkpour *et al.*, "Large Language Models Sensitivity to The Order of Options in Multiple-Choice Questions," in *Findings of NAACL*, 2024, pp. 2006–2017.

[27] E. J. Hu *et al.*, "LoRA: Low-Rank Adaptation of Large Language Models," in *ICLR*, 2022.

[28] T. Dettmers *et al.*, "QLoRA: Efficient Finetuning of Quantized LLMs," in *NeurIPS*, vol. 36, 2023, pp. 10 088–10 115.

[29] J. Chen *et al.*, "M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation," in *Findings of ACL*, 2024, pp. 2318–2335.

[30] N. F. Liu *et al.*, "Lost in the Middle: How Language Models Use Long Contexts," *TACL*, vol. 12, pp. 157–173, 2024.